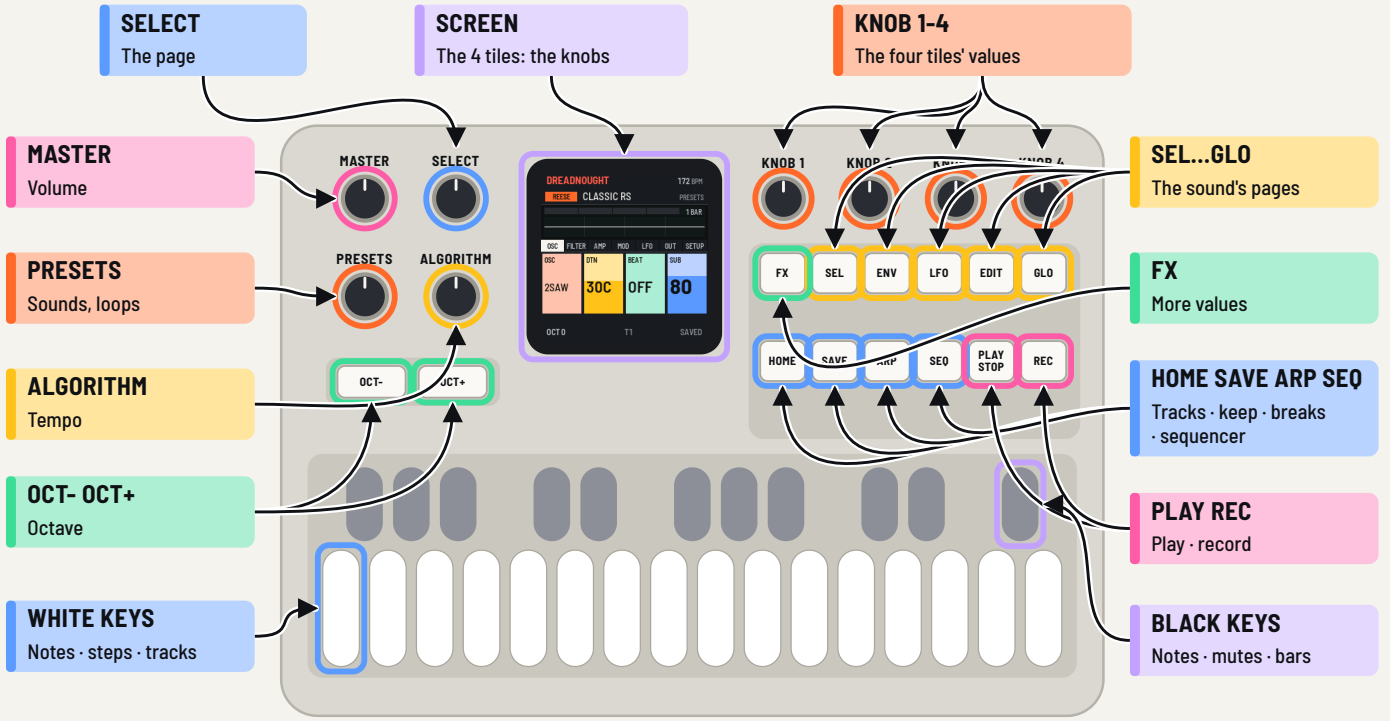
DREADNOUGHT

Dark jungle & drum and bass firmware for the M-VAVE FM-1

FIRMWARE	TRACKS	INSTRUMENTS	PATTERNS
0.33	8	8	64

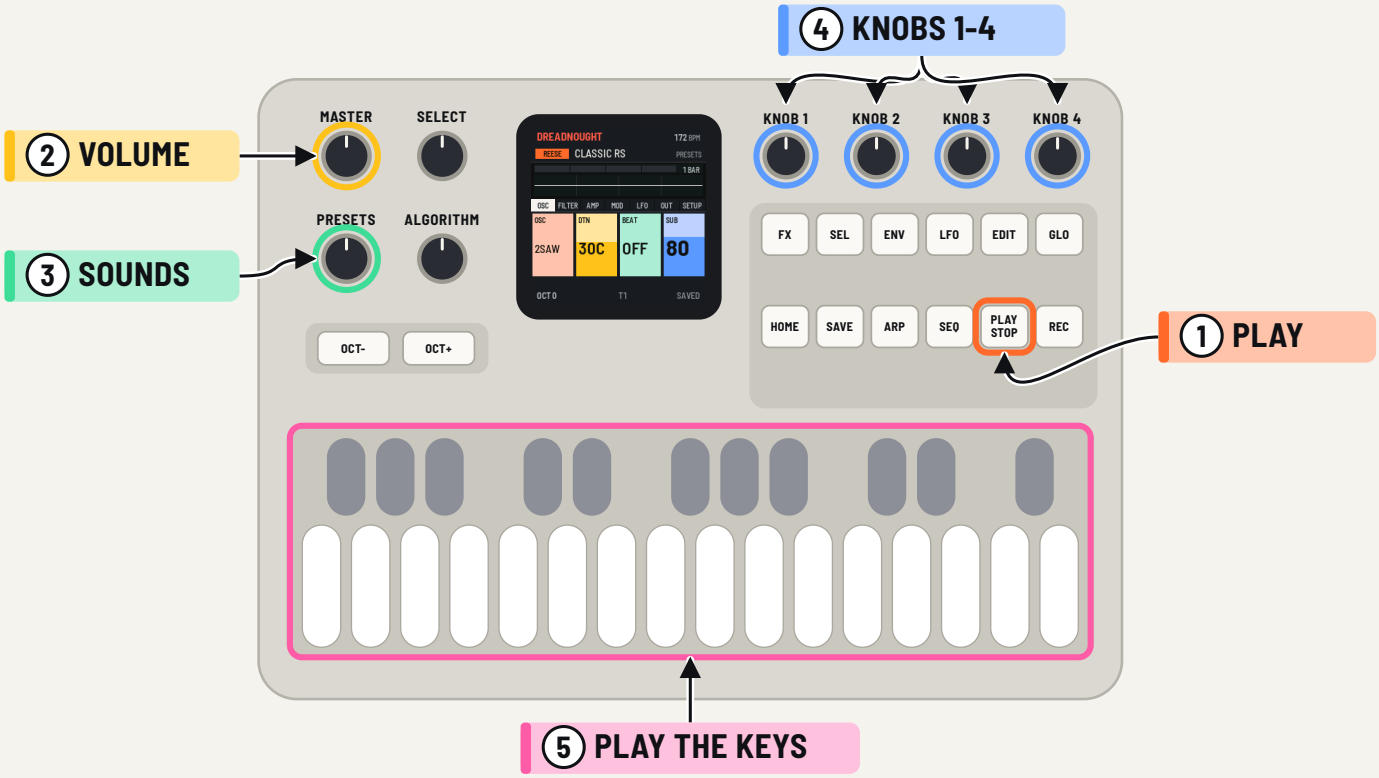


Manual & quick start · github.com/TwistedGenetics/DREADNOUGHT · a beta: back up your FM-1 first (see INSTALLING, UPDATING, RECOVERY)

QUICK START

Six walks round the box, each a few presses. Every one starts from the screen DREADNOUGHT starts on: REESE's OSC page. The numbers on the pictures are the order to press things in.

1 HEAR IT



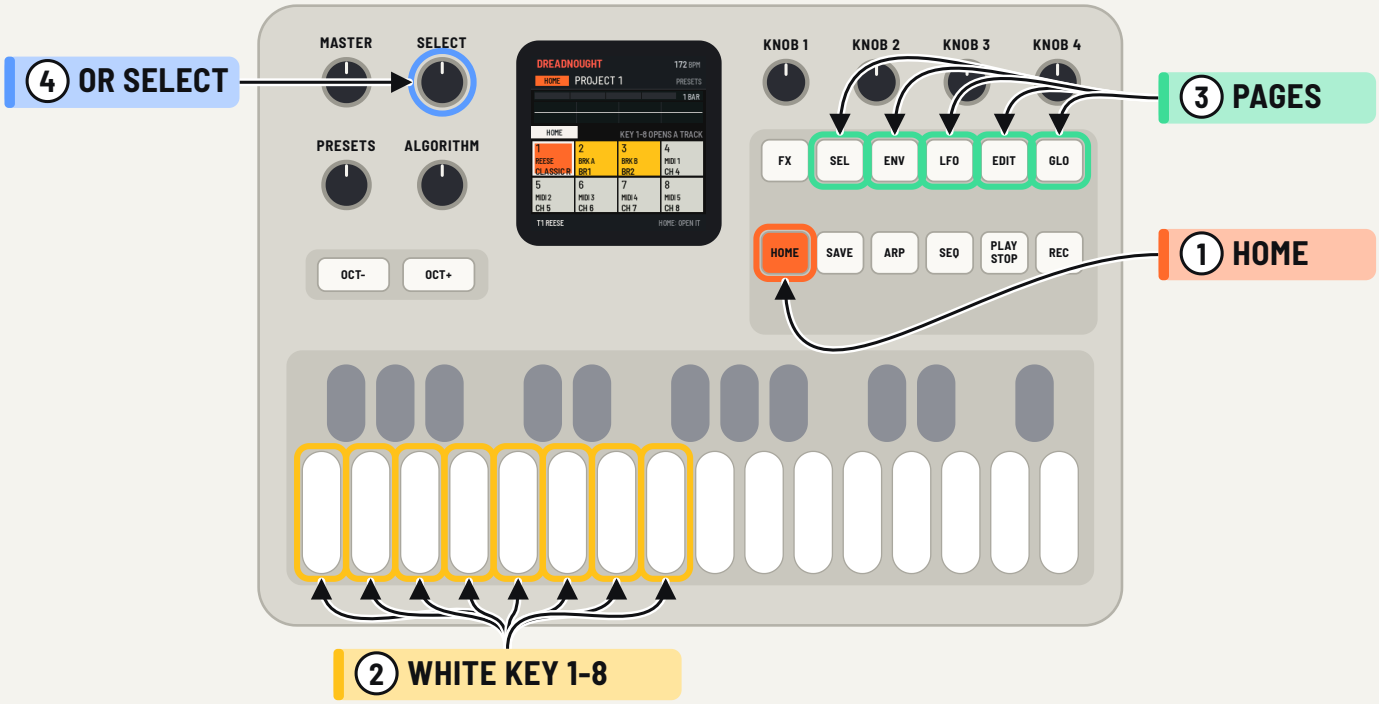
1. Press **PLAY**. The demo pattern plays: REESE, the bass on track 1. **PLAY** again stops it.
2. Turn **MASTER** for the volume.
3. Turn **PRESETS**: REESE's factory sounds, one a detent (CLASSIC RS, OG REESE, HOLLOW RS, TECHSTEP ...). The sound bar names it.
4. Turn **KNOB 1** – **KNOB 4**. Each changes the tile in its colour: orange, yellow, green, blue.
5. Play the keys: notes from F3 up to G5. Hold one and press another: it slides (REESE is mono).

ON THE SCREEN



REESE's OSC page: OSC, DTN, BEAT, SUB.

2 FIND YOUR WAY



1. Tap **HOME**. The eight tracks, each its number, instrument and sound, in its instrument's colour.
2. Press a white key 1-8 (the first eight from the left): that track's instrument opens on its first page.
3. **SEL** **EDIT** **ENV** **LFO** **GLO** are its pages: OSC, FILTER, AMP, MOD, OUT. Press one again for its second page (REESE: ENV again is the filter envelope).
4. Or turn **SELECT** to go through every page in turn. The tabs show where you are.

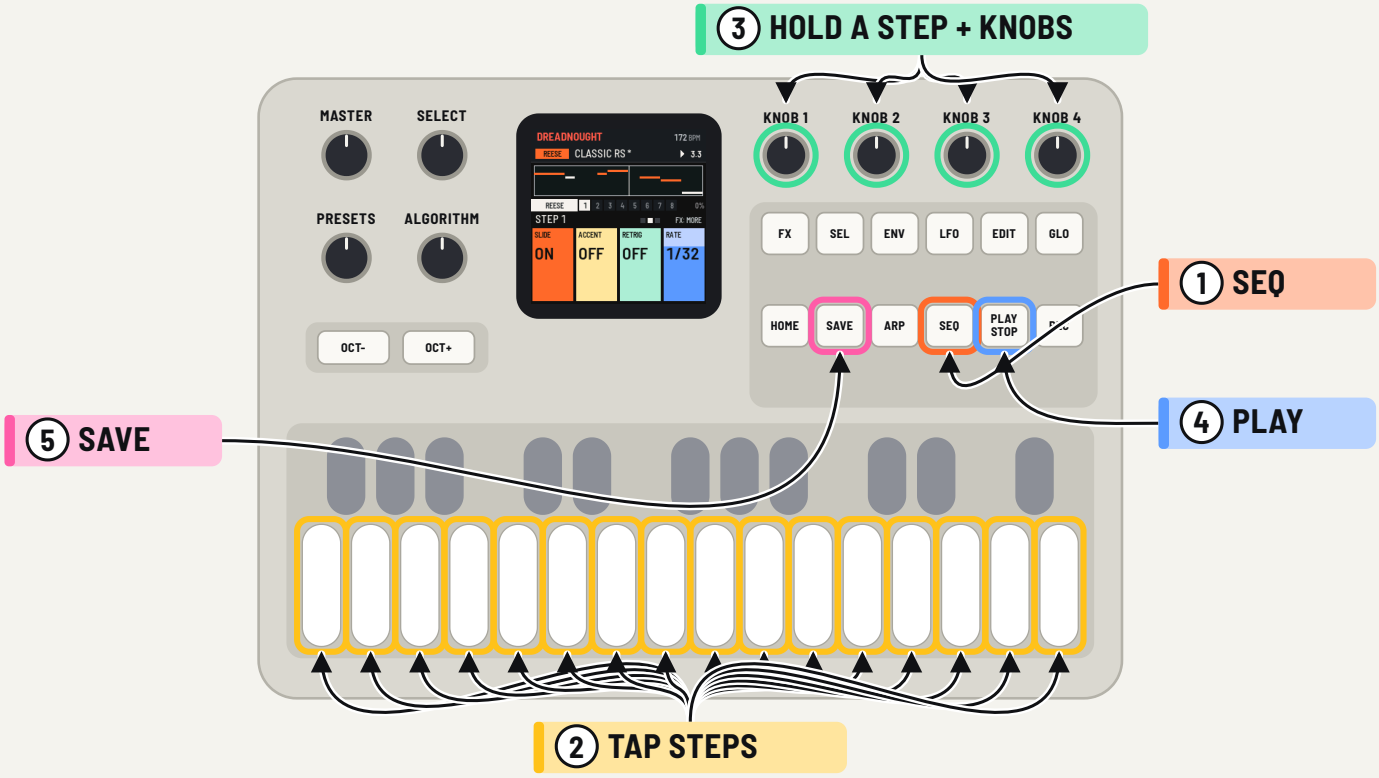
TIP From anywhere: hold **SEQ** and press a white key 1-8 to jump to that track's sequencer, or a black key 1-8 to its sound pages.

ON THE SCREEN



HOME: the eight tracks.

3 MAKE A PATTERN



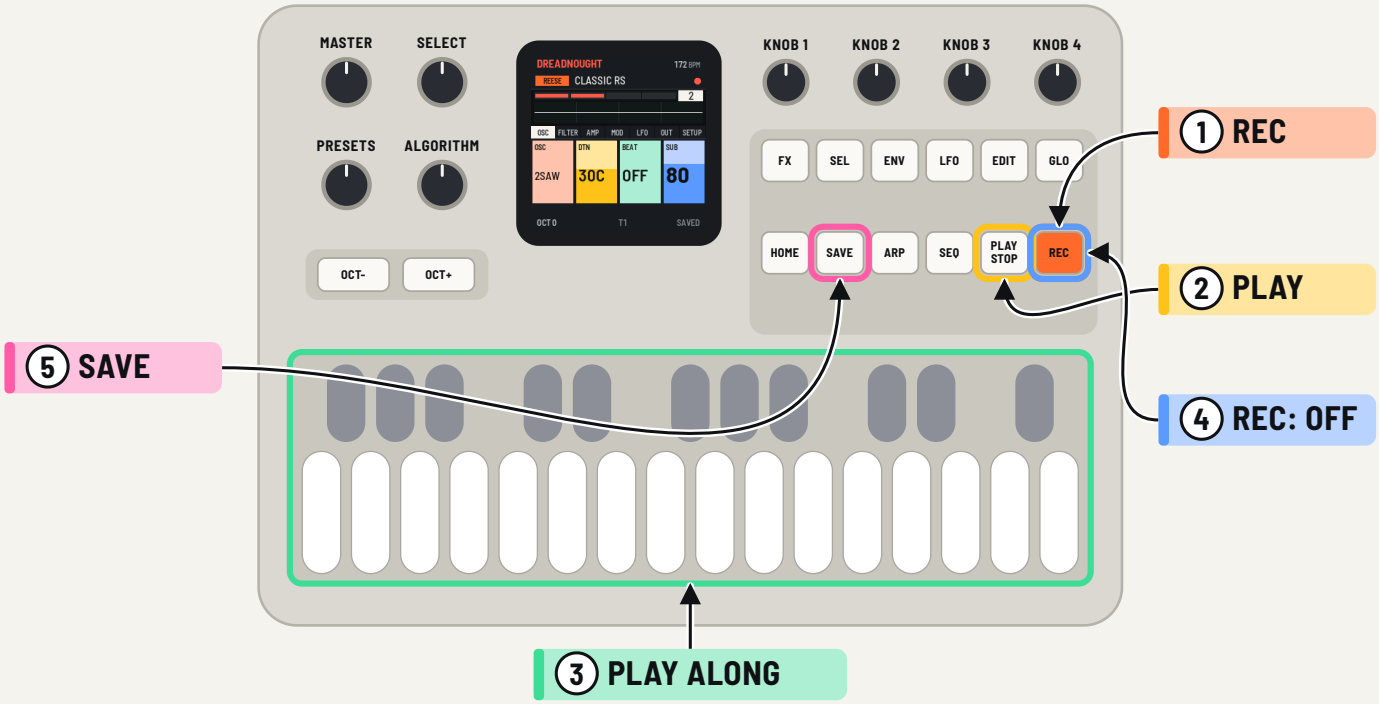
1. On the track you want, press **SEQ**: its piano roll. (On a break: the slice grid, DRUM.)
2. Tap white keys: steps 1-16 on and off. A new step gets the note, length and chord you last put in.
3. Hold a step: its four values pop up big. Turn **KNOB 1** NOTE, **KNOB 2** LENGTH, **KNOB 3** VEL, **KNOB 4** COND (a chance, FILL, every 2nd time...). **FX** while holding: SLIDE, ACCENT, RETRIG, RATE; again: the retrig's fade and CHORD.
4. **PLAY** to hear it. Black keys 10 and 11 show bars 2-4 (64 steps).
5. Tap **SAVE** (lit when something has changed) to keep it.

ON THE SCREEN



A step held: its values, big.

4 RECORD LIVE



1. Press **REC** (lit).
2. Press **PLAY**: a bar of clicks to count you in, then the pattern runs with a click on the beat.
3. Play: each note lands on its nearest step, as long as you held it. Keys held together on FM, WAVE, SAMPLER's keys or MIDI make a chord step.
4. Press **REC** again to stop recording; **PLAY** to stop.
5. Tap **SAVE**.

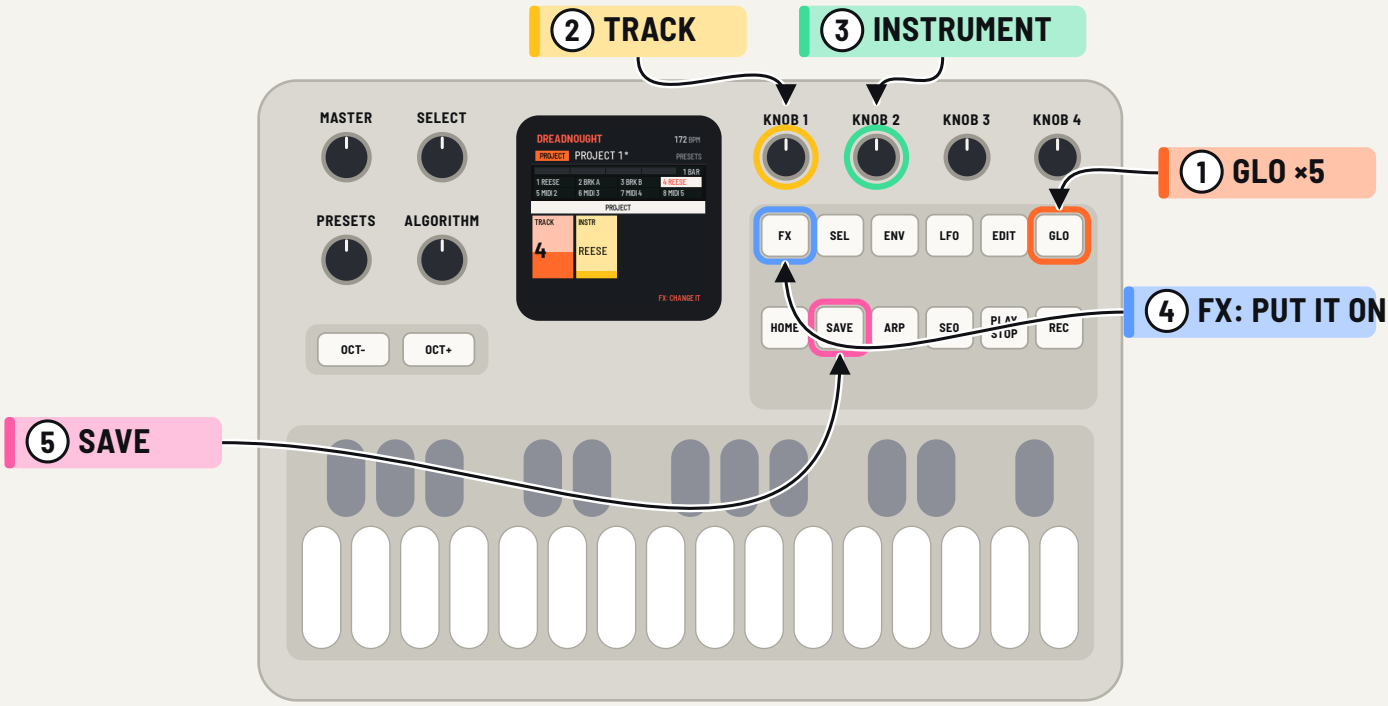
TIP Stopped, REC records step by step: each key writes the next step. The click and the count-in are on SETUP (**GLO** three times).

ON THE SCREEN



The count-in: the bar counts down.

5 CHOOSE THE INSTRUMENTS



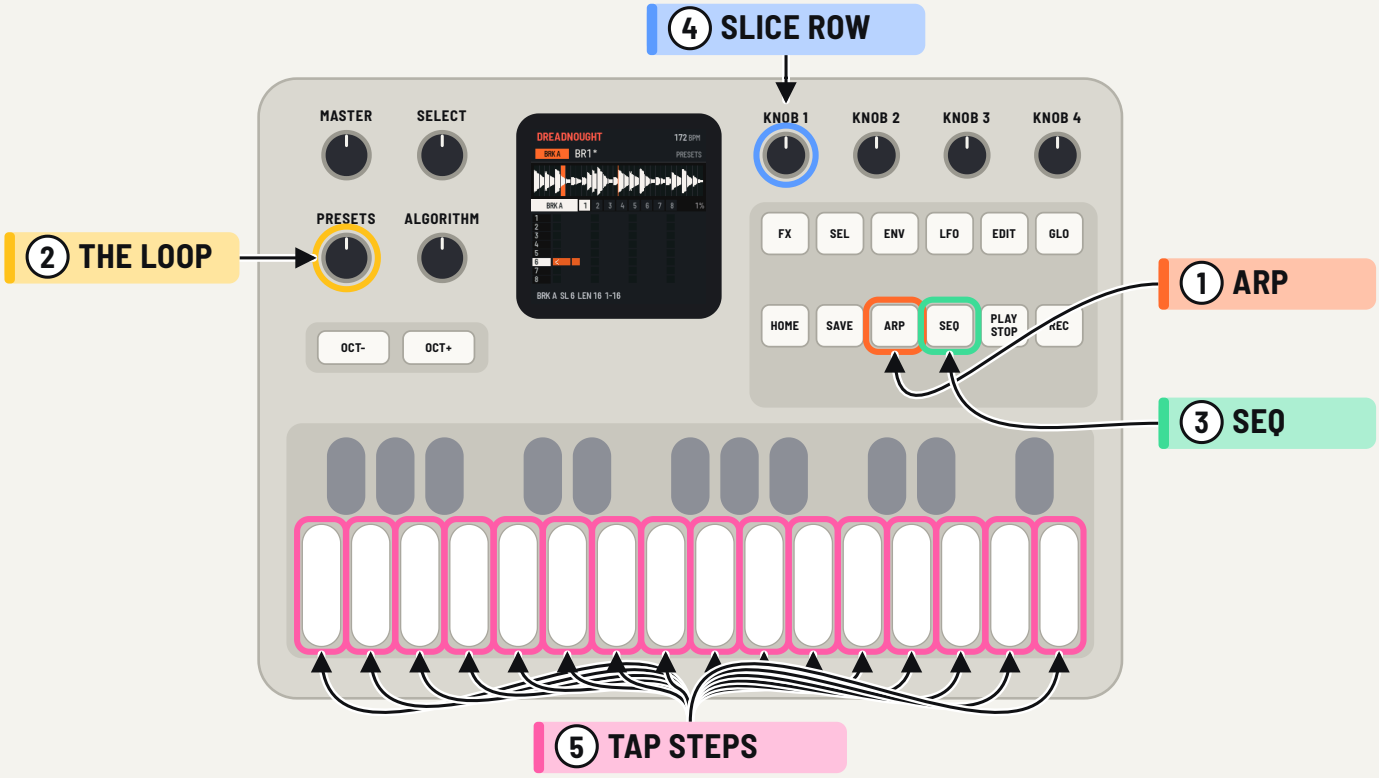
1. Press **GLO** five times: OUT, FX, SETUP, MIDI, PROJECT.
2. **KNOB 1**: the track.
3. **KNOB 2**: an instrument for it: OFF, REESE, BRK A, BRK B, MIDI, SUB, SAMPLER, FM or WAVE.
4. **FX** puts it on the track. REESE, SUB, SAMPLER, FM, WAVE and each break are on one track at a time: given to another, the two tracks swap. The patterns stay with their track.
5. **SAVE** keeps the layout.

ON THE SCREEN



PROJECT: a track and its instrument.

6 PLAY A BREAK



1. First send some loops: the breaks page or the web editor's Project tab (see THE WEB EDITOR). Until then the breaks are silent.
2. Open a break track (HOME, key 2) and press **ARP**: BRK, the loop's page. **ARP** again: BRK2, then SLICE.
3. Turn **PRESETS** to pick the loop.
4. **SEQ**: the slice grid. **KNOB 1** picks the slice row; tap white keys to put that slice on those steps.
5. **PLAY**. The loop follows the tempo (**ALGORITHM**).

ON THE SCREEN



DRUM: slices down, steps across.

1 THE PANEL

Everything on the FM-1, and what it does in DREADNOUGHT. The screen tells you what the four knobs do on each page; the buttons do the same thing everywhere.

CONTROL	DOES
MASTER	Volume.
SELECT	The page (on the sound pages). On the breaks' pages: BREAKS A / B. On MIDI: which MIDI instrument. With a step held: moves it.
PRESETS	The track instrument's sounds. On a break: the loop. With SAVE held: the user slot to store into. On ROLL with a step held: its chord.
ALGORITHM	The tempo, 60-200 BPM.
KNOB 1 – KNOB 4	The four tiles on the screen, in their colours: orange, yellow, green, blue.
OCT- OCT+	Octave, -3 to +3. With a step held: moves its note an octave.
SEL EDIT ENV LFO GLO	The sound's five pages (OSC, FILTER, AMP, MOD, OUT on the synths). Pressed again: the page's second view. GLO goes on: FX, SETUP, MIDI, PROJECT.
FX	More: the next four values of an effect, a held step, an LFO; on PROJECT, puts the instrument on the track; on SETUP, the skin.
HOME	Tap: the HOME page (the eight tracks). On a sequencer page, held: FILL.
SAVE	Tap: keep the sound, patterns, slices and layout (lit when something changed). Held with other controls: store, name, copy (see SAVING).
ARP	BREAKS: BRK, again BRK2, again SLICE (chop mode).
SEQ	This instrument's sequencer page; again: back. Held + a key 1-8: jump to a track.
PLAY	Start / stop the sequencer (MIDI clock, start and stop go out on USB).
REC	Record into the selected track: playing, live; stopped, step by step.
White keys	Notes. On ROLL and DRUM: the 16 steps in view. On HOME: tracks 1-8. On SLICE with REC lit: chops.
Black keys	Notes. On ROLL and DRUM: 1-8 mute tracks 1-8, 9 held 1 s clears the track, 10 / 11 the bar in view.
OCT- + OCT+ held 5 s	Update mode, for the flasher (recovery).

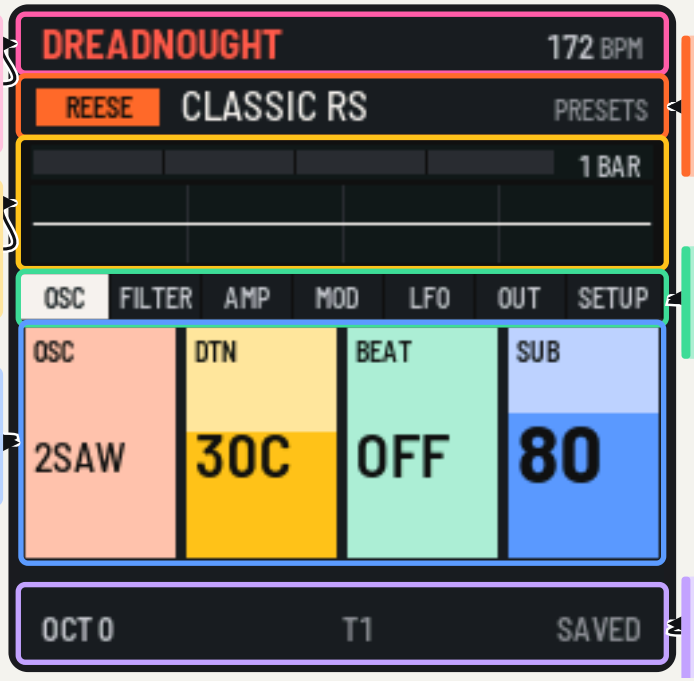
2 THE SCREEN

DREADNOUGHT's screen is built to be read from arm's length: big values, each knob its own coloured tile. This is the default skin, BLOCKS; SETUP has two more (see SETUP AND SKINS).

TOP BAR
DREADNOUGHT, the tempo; playing: the bar and beat.

BARS AND SCOPE
The pattern's bars, where it is, the beat; the sound's wave.

THE FOUR TILES
KNOB 1-4: name, value, filled as far as the value goes.



SOUND BAR
The instrument (REESE) and its sound. * changed since SAVE.

TABS
The sound's pages; lit: this one.

STATUS LINE
Octave, track, SAVED; messages.

The tile fills from the bottom as far as the value goes; a value that can go either way (a pitch, an envelope amount) fills from the middle. When something happens (a refusal, a pattern queued, "STORE TO USER 3") the status line says so for a moment.



HOME: the tracks in their instrument's colour.



A step held on ROLL: its values big.



PROJECT: the tracks and their instruments.

3 TRACKS, HOME AND PROJECTS

A project is eight tracks. Each plays one instrument and has 8 patterns of up to 64 steps.

Out of the box: track 1 REESE, tracks 2 and 3 BREAKS A and B, tracks 4-8 MIDI (your other gear, channels 4-8). PROJECT changes that (quick start 5); the instruments you can put on a track:

INSTRUMENT	WHAT IT IS
REESE	Detuned saws through a ladder filter: the jungle bass. Mono, slides.
SUB	A 101-style mono synth with a dirt stage: dread bass, acid, 808 subs.
BRK A, BRK B	Two break players: loops in time with the tempo, sliced and re-sequenced, chopped by hand.
SAMPLER	16 one-shot pads, or a key map of 8 zones (a multisample).
FM	A six-operator DX7-style synth (Dexed's engine): plays DX7 voices. 6 voices.
WAVE	A wavetable synth: 8 tables, your own Serum tables, unison, warps. 6 voices.
MIDI	Plays other gear on its channel over USB. Any number of tracks.
OFF	Plays nothing.

HOME

Tap HOME from any sound page. Each track shows its number, instrument and sound; a dot lights when it plays a note, a muted track is dim. A white or black key 1-8 opens that track. Or turn SELECT to pick one and tap HOME again. Hold HOME and turn SELECT to step through the instruments without leaving the page you're on.

JUMPING

SEQ **held + white key 1-8**	That track's instrument (on a sequencer page: its sequencer page).
SEQ **held + black key 1-8**	That track's sound pages, from anywhere.
HOME **held +** SELECT	The next / previous instrument's pages.

SAVING

SAVE keeps everything: each instrument's sound, all 64 patterns, the layout, the effects, your slices. It comes back at power-on. The project has a name: on PROJECT, hold SAVE and press FX to name it (see SAVING). The web editor saves whole projects to your computer.

4 REESE

Up to seven drifting saws, a Moog-style ladder with drive, a swept comb for the hollow movement and a clean sine on the fundamental. Mono: a key held under another slides.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
OSC	SEL	OSC: 2SAW … 7SAW, SAWSQ, 2SIN	DTN: detune	BEAT: the movement locked to the tempo	SUB: the sine
FILTER	EDIT	CUT	RES	DRV: drive	NTCH: the comb
AMP	ENV	ATK	DEC	SUS	REL
F.ENV	ENV again	F.ATK	F.DEC	F.SUS	F.REL
MOD	LFO	FENV: filter env amount	GLIDE (ms)	L1 RATE	L1 DEPTH
OUT	GLO	LEVEL	TRANS	BPM	OVERLAP: SLIDE / CUT

PRESETS: CLASSIC RS, OG REESE, HOLLOW RS, TECHSTEP, NEURO RS, SYNC RS, DARK RS, SUB REESE, HOOVER RS, INIT, then USER 1-8. FX on AMP: ENV SYNC, the envelope's times as note lengths at the tempo.



OSC.



FILTER.



F.ENV (ENV again).

TIP OVERLAP: SLIDE glides a note into the next one when its length reaches past it (no new attack): the classic rolling reese. CUT retriggers every note.

5 SUB

A 101-style mono synth: saw, pulse with PWM, a sub-oscillator and noise into a resonant filter that tracks the keys, then the dirt.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
OSC	SEL	SAW	PULSE	SUB	NOISE
OSC2	SEL again	PW	PWM	PWM RATE	SUB MODE: -1 OCT / -2 OCT / -2 25%
FILTER	EDIT	CUT	RES (self-oscillates)	ENV	KEY: tracking
DIRT	EDIT again	DRIVE	TYPE: SOFT / HARD / FOLD / RECT	CRUSH	TONE
AMP	ENV	ATK	DEC	SUS	REL
F.ENV	ENV again	F.ATK	F.DEC	F.SUS	F.REL
MOD	LFO	GLIDE	PORTA: OFF / AUTO / ON	ACCENT	VCA: ENV / GATE
OUT	GLO	LEVEL	TRANS	SPLIT: below it, no dirt	OVERLAP

PRESETS: DREAD, DONK, ROLLER, WOBBLE, ACID 101, 808 SUB, GROWL, SQUARE SB, PUNCH, TECH SUB, CRUSHED, RAVE STAB, INIT, then USER 1-8. ACCENT plays the sequencer's ACCENT steps louder and brighter, as the 303 does.



OSC.



DIRT (EDIT again).



MOD.

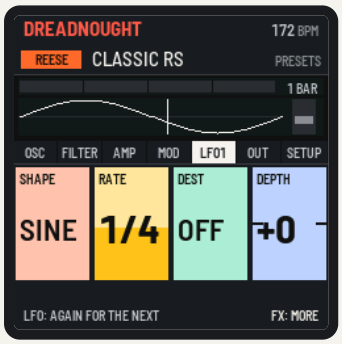
6 LFOS

REESE and SUB each have four LFOs, part of their sound: a preset brings its own.

Press LFO on REESE or SUB: MOD. Again: LFO 1, then 2, 3, 4, then back to MOD. FX shows the LFO's MORE values, and on a DRAW LFO, the drawing.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
LFO 1-4	LFO again	SHAPE: SINE TRI SAW UP SAW DN SQUARE S&H SMOOTH DRAW	RATE: Hz, or a note length	DEST: where it goes	DEPTH: + or -
MORE	FX	SYNC: FREE / SYNC	MODE: FREE / RETRIG / 1-SHOT	PHASE	SMOOTH
DRAW	FX again	the pen (a point's value)	SMOOTH	POINTS: 16 / 32	points in view

DEST: the synth's own values (CUTOFF, RES, DRIVE, NOTCH, DETUNE, PITCH, SUB, LEVEL, F.ENV; SUB's S CUT, S RES ...), the breaks' (A PITCH, A DECAY ...) or another LFO's depth. DRAW: the white keys are the points; press one to set it to the pen.



An LFO.



MORE (FX).



DRAW: a shape by hand.

7 BREAKS

Two break players, A and B, on tracks 2 and 3 unless you move them. They play the loops you send from the editor or the breaks page, always in time.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
BRK	ARP	LOOP (PRESETS too)	PITCH	DECAY (127: hold)	CRUSH
BRK2	ARP again	CHOKE: group OFF, 1-4	SLICES: GRID / AUTO / MANUAL	STRETCH: REPITCH / SLICE / STRETCH	LEVEL
SLICE	ARP again	CHOP: select	CURSOR (coarse)	FINE	ZOOM

SELECT on these pages: BREAKS A or B. A and B choke each other (group 1) like one sampler voice.



BRK.



DRUM, playing.



SLICE: chops.

THE SLICE GRID (DRUM)

SEQ from a break's page. Each row is a slice, each column a step. KNOB 1 picks the row; tap white keys to put it on steps. KNOB 2 the track's length, KNOB 3 swing, KNOB 4 the steps in view. Hold a step: SLIDE plays it reversed. SAVE + FX: the break laid out straight, each step its own slice.

CHOP MODE (SLICE)

Set SLICES to MANUAL on BRK2, then ARP again. The MPC way: press PLAY, press REC (lit: the keys cut) and hit the lowest key on the first hit, the next key on the next, and so on. Each chop lands where you hit it; hit a key again to retake it. REC off: the keys play the chops. Chops save themselves.

ON SLICE	DOES
KNOB 1	Select a chop (the cursor jumps there, you hear it).
KNOB 2 / KNOB 3	The cursor: coarse / fine.
KNOB 4	Zoom: the whole loop ... 1/64 of it.
FX	Move the selected chop to the cursor.

ON SLICE	DOES
HOME	Delete the selected chop; held 1 s: delete them all.
SAVE + REC	A chop between the others, at the cursor.
SAVE + OCT- / OCT+	Start again from AUTO's hits / from the GRID.

8 SAMPLER

Your samples (sent with the editor) on 16 pads, or mapped across the keys.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
PAD	SEL	SAMPLE	START	END	REVERSE
FILTER	EDIT	CUT	RES	DRIVE	CRUSH
AMP	ENV	ATTACK	HOLD (127: to the end)	DECAY	LEVEL
MOD	LFO	PITCH	FINE	PAN	CHOKE
OUT	GLO	MODE: ONE-SHOT / KEYS	PAD	BPM	

ONE-SHOT: 16 pads, played by the first 16 keys and sequenced on DRUM's rows. PRESETS: the pad shown.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
ZONE	SEL	SAMPLE	ROOT	LOW	HIGH
FILTER	EDIT	CUT	RES	DRIVE	CRUSH
AMP	ENV	ATTACK	DECAY	SUSTAIN	RELEASE
LOOP	LFO	LOOP: OFF / LOOP / SUST	L.START	L.END	TUNE
OUT	GLO	MODE	ZONE	LEVEL	VOICES (1: mono)

KEYS: 8 zones, each a sample over a range of keys. Sequenced on ROLL; plays chords.



ONE-SHOT: a pad.



The pads on DRUM.



KEYS: a zone.

9 FM

A six-operator FM synth that plays DX7 voices: ten factory voices, and any you keep from a DX7 bank in the editor.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
VOICE	SEL	ALGO 1-32	FEEDBACK	BRIGHT: the modulators	TIME: every envelope
OP	EDIT	OP: which of the six	COARSE (ratio / Hz)	FINE	LEVEL
OP+	EDIT again	OP	DETUNE	VEL	MODE: RATIO / FIXED
EG	ENV	R1	R2	R3	R4
EG LVL	ENV again	L1	L2	L3	L4
LFO	LFO	SPEED	DELAY	PMD	AMD
LFO+	LFO again	WAVE	SYNC	PMS	KEY SYNC
OUT	GLO	LEVEL	VOICES 1-6 (1: mono)	ATTACK	RELEASE

PRESETS: DUB FM BS, FM REESE, HOLLOW STB, DARK BELL, METAL HIT, FM PLUCK, DARK PAD, GROWL BASS, SINE SUB, INIT VOICE, then your DX7 voices. BRIGHT, TIME, ATTACK and RELEASE sit on top of the voice without changing it.



VOICE.



OP: the operator's frequency.



EG: OP3's rates.

10 WAVE

A wavetable synth with two oscillators (0.33): 8 factory tables and your own Serum tables, A and B, FM, ring, AM and sync between them, warps, six filters, seven drives, three envelopes, three LFOs and a matrix of 8 routes.

Each page button pressed again shows the page's next view; the tab names it. SEL: WAVE A, OSC A, WAVE B, OSC B, B MIX. EDIT: FILTER, DRIVE. ENV: AMP, MOD ENV, ENV 3. LFO: MOD, LFO 1, LFO 2, LFO 3, MATRIX.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
WAVE A	SEL	TABLE	POS: through the frames	WARP: OFF SYNC BEND MIRROR PWM FM CRUSH	AMOUNT
OSC A	SEL ×2	UNISON 1-4	DETUNE	SPREAD: stereo	SUB: a sine an octave down
WAVE B	SEL ×3	B TABLE	B POS	B WARP	B AMT
OSC B	SEL ×4	OSC B: OFF / ON	B OCT	B SEMI	B FINE
B MIX	SEL ×5	B LEVEL: B heard	B MODE: MIX FM RING AM SYNC	MOD AMT	NOISE
FILTER	EDIT	FILTER: LP12 LP24 HP BP NOTCH COMB	CUT	RES	ENV: mod env on the cutoff
DRIVE	EDIT ×2	DIST: SOFT HARD FOLD RECT TUBE BITS RATE	DRIVE	SAT: after the filter	KEY: cutoff follows the note
AMP	ENV	ATK	DEC	SUS	REL
MOD ENV	ENV ×2	M.ATK	M.DEC	M.SUS	M.REL
ENV 3	ENV ×3	E3 ATK	E3 DEC	E3 SUS	E3 REL
MOD	LFO	POS ENV	POS LFO	L1 RATE	CUT LFO
LFO 1-3	LFO ×2-4	SHAPE: SINE TRI SAW SQUARE S&H RANDOM	RATE: Hz, or a length	SYNC: FREE / SYNC	MODE: FREE / RETRIG
MATRIX	LFO ×5	SLOT 1-8	SOURCE	DEST	DEPTH
OUT	GLO	LEVEL	VOICES 1-6 (1: MONO)	GLIDE	TRANS

PRESETS: INIT, SUPERSAW, REESE, PPG BELL, VOX PAD, SYNC LEAD, PWM STRNG, GRIT BASS, ORGAN, FM GROWL, BENT KEYS, CRUSHER, FM BELL, SYNC SWEEP, RING DRONE, COMB PLUCK, FOLD BASS, HOOVER, BIT WOBBLE, S&H

TWIST.



WAVE A.



B MIX (SEL ×5).



DRIVE (EDIT again).



MATRIX (LFO ×5).

B MODE	WHAT IT DOES
MIX	B is only heard, at B LEVEL.
FM	B moves A's phase: MOD AMT is how far. Bells, growls, metal. B LEVEL 0 to hear only A.
RING	A times B: their sum and difference, A's own pitch gone at MOD AMT 127. Clangs and drones.
AM	B moves A's level: tremolo at low B pitches, sidebands at high ones.
SYNC	A starts again with each of B's cycles. A plays MOD AMT above B (up to 4 octaves): sweep MOD AMT (the matrix: MOD ENV to MOD AMT) for the classic sync scream.

MATRIX	
Sources	LFO 1-3, MOD ENV, ENV 3, AMP ENV, VEL (velocity), KEY (the note), RANDOM (one per note).
Destinations	POS A, POS B, WARP A, WARP B (their AMOUNTs), PITCH, PITCH B, MOD AMT, B LEVEL, CUTOFF, RES, DRIVE, LEVEL, PAN, SUB, NOISE, L1 RATE, L2 RATE, DETUNE.
Depth	-64 to +63: how far, either way. Routes add up.

TABLE	POS 0 → 127
ANALOG	Sine, to triangle, to square, to saw.
REESE	A saw with notches that move: two detuned saws, frozen.
PPG	A band of harmonics stepping up, on a thin saw.
VOCAL	The vowels A E I O U, one into the next.
SYNC	Hard sync: a saw restarting 1 to 8 times a cycle.
PWM	A pulse from 50 % down to 4 %.
GRIT	A sine folded harder and harder.

TABLE	POS 0 → 127
HOLLOW	Odd harmonics, drawbars moving: an organ going hollow.

VOICES The voices WAVE plays depend on what's switched on, so the FM-1 never runs out of time: UNISON, OSC B, a DIST, LP24, SAT and stereo (SPREAD, PAN) each cost some. Plain: 6 voices. UNISON 2: 3, UNISON 4: 2. OSC B on: 3, and unison 2 at most. B on with a drive: 2. VOICES is the most it will use; watch CPU on PROJECT.

11 MIDI TRACKS

Tracks set to MIDI play your other gear over USB, each on its channel.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
MIDI	GLO ×4	MIDI: which (1-5…)	CHANNEL 1-16	OVERLAP: SLIDE / CUT	

SELECT on MIDI: the next MIDI instrument. With a MIDI track selected the keys play out on its channel. SLIDE sends the new note before the old one's note-off: legato on the other synth.

12 THE SEQUENCER

Every instrument has its own sequencer page: ROLL (a piano roll) for the synths, SAMPLER's keys and MIDI; DRUM (a slice grid) for the breaks and SAMPLER's pads.



ROLL: REESE's pattern.



Marks: conditions, retrigs, accents.



DRUM: a break.

ON ROLL / DRUM	DOES
White keys	The 16 steps in view: tap on / off.
Black keys 1-8	Mute / unmute tracks 1-8 (lit: playing).
Black key 9, held 1 s	Clear the track.
Black keys 10 / 11	The bar in view: steps 1-16, 17-32, 33-48, 49-64.
KNOB 1 – KNOB 4 (ROLL)	Scroll the pitches, transpose the track, its length (1-64), the steps in view.
KNOB 1 – KNOB 4 (DRUM)	The slice row, length, swing, the steps in view.
FX (ROLL)	KEYS mode: the keys write notes at a cursor. KNOB 2 their length, KNOB 4 the cursor, HOME a rest.
HOME held	FILL, for FILL / !FILL steps.

A STEP HELD

Hold a step's key: its values pop up big, four tiles, one a knob. FX (still holding) shows the next four.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
1	hold a step	NOTE (break: slice)	LENGTH	VEL	COND
2	FX	SLIDE (break: reverse)	ACCENT	RETRIG	RATE
3	FX again	RT LEN	RT VEL: fade in / out	CHORD	

DOING	HOW
Move a note	Hold its step and press the step it should go to (black keys 10 / 11: another bar; onto a note: they swap). Or turn SELECT. OCT- / OCT+: an octave.
Copy a note	As moving, with SAVE held.
Velocity ramp	Hold the first step, hold REC, press the last; KNOB 3 / 4 the ramp's start and end.
Retrigs	RETRIG on: the step repeats at RATE (1/16 a step, 1/32 two …) for RT LEN, its velocity fading (RT VEL): rolls and builds.
Chords	FM, WAVE, SAMPLER's keys and MIDI: 15 chords (MIN, MAJ, MIN7, MAJ7, 7, MIN9, MAJ9, ADD9, SUS2, SUS4, 5, DIM, AUG, MIN6, 6). Hold a step and turn PRESETS, or CHORD on page 3, or record keys held together.
Conditions	COND: a % chance; FILL / !FILL; PRE / !PRE (the track's last condition); NEI / !NEI (the track before's); 1ST / !1ST; LST / !LST; A:B (the A-th of every B passes).
Overlaps	A note whose length reaches the next slides into it (OVERLAP SLIDE on OUT) or retriggers it (CUT).

PATTERNS

Each instrument has 8 patterns, along the top of its sequencer page: the playing one lit, a queued one flashing, ones with notes marked, and how full pattern memory is.

COMBINATION	DOES
SAVE + white key 1-8	This instrument's pattern 1-8: stopped, at once; playing, when the current one ends.
SAVE + black key 10 / 11	Copy this pattern / paste into this one.
SAVE + OCT- / OCT+	Copy / paste the bar in view.
SAVE + REC	×2: the bars with notes copied into the empty bars after them; a full track doubles (16 → 32 → 64 steps).

RECORDING

REC lit, playing: each key writes a note at the nearest step, as long as you hold it. Stopped: step by step from the cursor; a key held under another writes a slide. Keys held together make a chord step on the instruments that play chords. A new step you tap in takes the last note, length and chord you used on that track.

13 EFFECTS

Two effect slots, FX A and FX B, fed by each track's sends. A master limiter keeps the mix from clipping.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
MIX	GLO ×2	SEND A (the track)	SEND B	RETURN A	RETURN B
FX A	FX	the effect	its 1st value	2nd	3rd
FX A 2	FX again	its 4th	5th	6th	7th
FX B, FX B 2	FX again	as FX A			

176 KB of effect memory: an effect that won't fit beside the other is refused. DELAY fits beside anything; two of the big reverbs fit; two DELAYs don't.



MIX: the sends and returns.



FX A: MASSIVE.



QUADRA, FLANGE.

EFFECT	WHAT IT IS	VALUES
DELAY	The dark half-rate ping-pong.	TIME (synced), FEEDBACK, TONE
TAPE	Tape echo: wow, flutter, head loss, saturation.	TIME, FEEDBACK, DEGRADE, PINGPONG
BBD	Bucket brigade: crumbly, dark.	as TAPE
SPRING	Two springs: the dub drip.	TIME, DAMP, PREDELAY, LOW CUT
TOMB	A small stone room.	as SPRING
MOD	A big moving hall.	as SPRING
CELESTIAL	MOD with an octave-up shimmer.	as SPRING
DOLMEN	A small dense chamber, bright.	as SPRING
TUMULUS	A cavern: long and dark.	as SPRING

EFFECT	WHAT IT IS	VALUES
QUADRA	Quadraverb-style chains: CHORUS, REVERB, 5EQ>REV, FLANGE, DETUNE, DLY>REV.	CONFIG, DECAY, TIME, FEEDBACK, MOD, TONE, 16-BIT
MASSIVE	Super-Massive: ROOM, HALL, HALO, SWELL, ABYSS, ECHOVRB, ORBIT, PRISM, PONG, DUB.	MODE, TIME, FEEDBACK, WARP, DENSITY, MOD, TONE

14 SETUP AND SKINS

GLO three times.

PAGE	GET THERE	KNOB 1	KNOB 2	KNOB 3	KNOB 4
SETUP	GLO ×3	CLICK: OFF / REC / ON	CLK LVL	COUNT-IN: off / 1 / 2 bars	VEL SENS (REESE)

FX on SETUP changes the skin: BLOCKS (the default: big coloured tiles), LIGHT (warm white, ink, orange), DARK (the first look). SAVE keeps it.



BLOCKS.



LIGHT.



DARK.

15 SAVING, USER SOUNDS, NAMES

DOING	HOW
SAVE (tap)	Keep everything for next time. Lit when something changed.
SAVE held + PRESETS	Store REESE's or SUB's sound (with its LFOs) in USER 1-8: turn to pick, let go of SAVE to store. The name editor opens.
SAVE held + FX	Name it: on PROJECT the project, on a synth the user sound loaded.
The name editor	KNOB 1 or SELECT the place, KNOB 2 or PRESETS the letter; SAVE keeps it, HOME leaves it as it was.

TIP FM and WAVE keep their sound with SAVE too; FM's extra voices and WAVE's extra tables live in the sample slots, sent from the editor.

16 THE WEB EDITOR AND THE BREAKS PAGE

In Chrome or Edge, with the FM-1 on USB: twistedgenetics.github.io/DREADNOUGHT/editor

The editor opens on a big copy of the FM-1: its screen, knobs, buttons and keys. Everything there does what it does on the FM-1, and turning a knob on the FM-1 moves the page. The tabs after it edit in depth:

TAB	WHAT'S THERE
Pattern	Every track in a mouse-driven piano roll or slice grid; velocity, conditions, retrigs, chords, the 8 patterns.
REESE, SUB, LFOs	Every value; LFO shapes drawn with the mouse.
SAMPLER	The 16 pads or the key map.
FM	Every value of the voice. DX7 banks: load a .syx, hear each voice here or on the FM-1, keep some, send them.
WAVE	Every value. Wavetables: load a Serum wavetable or single-cycle WAV, see its frames, hear it, send it: it comes up on TABLE.
Breaks	The breaks' settings and slices.
Project	Each track's instrument and sends, the FX, memory and CPU; samples (WAV, MP3, AIFF ... converted and sent); save and load whole projects.

Sample memory: three slots, 80 + 80 + 64 KB. Samples, loops, DX7 voices and wavetables share it. A wavetable is about 2 KB a frame. The editor sends only what changed.

The breaks page (twistedgenetics.github.io/DREADNOUGHT/breaks) sends loops straight to the slots and sets their slices.

> **TIP** If the editor won't connect: close everything else using the FM-1's MIDI port (a DAW, the flasher or the breaks page in another tab), reload, Connect again. Diagnostics at the bottom of the page lists the ports.

17 MEMORY AND CPU

Patterns	8 per instrument, 64 steps each; notes of all 64 patterns share memory (about 1 680). The sequencer page shows how full it is.
Samples	224 KB in three slots, shared by loops, samples, DX7 voices and wavetables.
Effects	176 KB between FX A and FX B.

CPU	PROJECT shows the load and its peak (CPU 40 PK 62%). Too much at once: the status line says AUDIO LATE: CPU FULL. Fewer voices, less unison or a smaller reverb bring it back.

18 INSTALLING, UPDATING, RECOVERY

1. Open twistedgenetics.github.io/DREADNOUGHT/flasher in Chrome or Edge, the FM-1 on USB.
2. Back up first: the flasher saves your FM-1's own firmware.
3. Pick the newest release (or a .fwsc file from GitHub Releases) and press Install. The FM-1 restarts on DREADNOUGHT's screen.

RECOVERY If something goes wrong: hold OCT- and OCT+ together for 5 seconds. The FM-1 goes into update mode and the flasher can install again, or put the stock firmware back with M-VAVE's tools. docs/RECOVERY.md on GitHub has every way back. Updates keep your sounds, patterns and samples.

19 QUICK REFERENCE

The combinations, on one page.

PRESS	FOR
HOME	the eight tracks; key 1-8 opens one
SEQ	this instrument's sequencer; again: back
SEQ + white 1-8	that track's sequencer / instrument
SEQ + black 1-8	that track's sound pages
HOME + SELECT	next / previous instrument
SEL ... GLO again	the page's second view
GLO ×2 / ×3 / ×4 / ×5	FX · SETUP · MIDI · PROJECT
ARP ×1 / ×2 / ×3	BRK · BRK2 · SLICE
LFO again (REESE, SUB)	LFO 1 ... 4
step held + knobs	NOTE LENGTH VEL COND; FX: SLIDE ACCENT RETRIG RATE; FX: RT LEN RT VEL CHORD
step held + another step	move it there (SAVE held: copy)
step held + REC + last step	velocity ramp
SAVE + PRESETS	store to USER 1-8
SAVE + FX	name the project / sound
SAVE + white 1-8	pattern 1-8
SAVE + black 10 / 11	copy / paste the pattern
SAVE + OCT- / OCT+	copy / paste the bar
SAVE + REC	×2: double the pattern
HOME held (sequencer)	FILL
black 9 held 1 s	clear the track
OCT- + OCT+ 5 s	update mode